

Sniffing

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Discipline: Offensive and Defensive Cybersecurity

Lecturer: Carlos Antunes

Team: João Marto

Table of Contents:

1.Introduction and Framework	3
1.1.Laboratory Objectives	3
1.2.Scope and Scenario of Tests	4
2.Topology and Capture Preparation	5
2.1.Network Status Check and ARP	5
3.Running Traffic Capture (Sniffing)	6
3.1.Forensic Capture with Tcpdump	6
3.2.Unified Listening with Ettercap	6
4.Analysis of Captured Information	8
4.1.Artifact Mining with NetworkMiner	8
5.Conclusions and Mitigation Measures.....	10
6.Command Log	11

1. Introduction and Framework

This report documents the execution of **Laboratory 5 – Sniffing**, within the scope of the curricular unit of Offensive and Defensive Cybersecurity. The primary objective of this hands-on activity is to familiarize yourself with *sniffing* tools and techniques in a real network environment.

Sniffing is a key technique for both system administrators (in diagnosing network problems) and for security auditors and attackers (in intercepting sensitive information, credentials and communications mapping).

1.1. Laboratory Objectives

According to the script of Practical Fact Sheet 5, the tasks proposed for this report focus on three fundamental pillars:

- Perform traffic capture adapted to the installed network structure.
- Identify and analyze in detail the intercepted information.
- Prepare this report supported by the forensic evidence collected.

1.2. Scope and Scenario of Tests

To comply with the requirements, the tests were conducted on the author's local area network (LAN) infrastructure. As it is a switched network, traffic capture requires specific techniques to ensure that packets that are not destined directly to the auditor's machine can be intercepted, and the capabilities of the interfaces and broadcast/multicast traffic analysis are exploited in a promiscuous way. All procedures adopted an ethical stance, limiting listening to controlled equipment in the laboratory environment.

2. Topology and Capture Preparation

Performing *sniffing* on a local network requires proper preparation of the environment to ensure that traffic is captured in integrity and without disrupting existing infrastructure.

2.1. Network Status Check and ARP

Before starting any capture, the ARP table of the audit machine was checked. This practice is essential to record the "clean" state of the network and confirm the mappings between the IP addresses and the physical (MAC) addresses of the *Gateway* and the other *hosts*, serving as a baseline.

Command used (ARP Table Check):



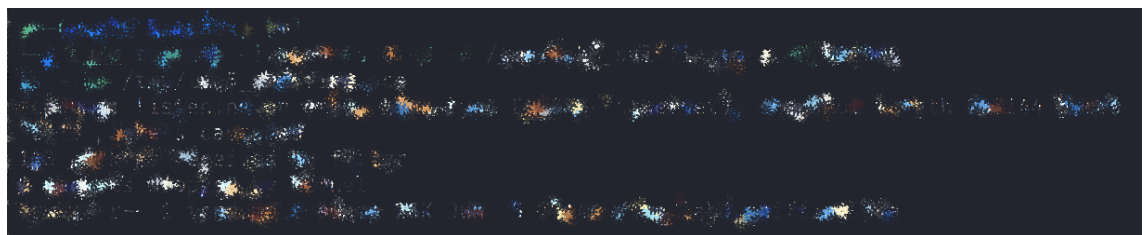
3. Running Traffic Capture (Sniffing)

To meet the requirement of capturing traffic according to the structure of the installed network, a mixed approach was used: file-focused capture for forensic evidence preservation and active listening with a graphical interface.

3.1. Forensic Capture with Tcpdump

The tcpdump utility configured to capture full-size packets (-s 0) on the main interface (eth0) was used. To facilitate the analysis process, all raw traffic was directed to the system's temporary folder, creating an evidence file (.pca).

Command used (File Capture and Verification):



3.2. Unified Listening with Ettercap

As a complement to command-line capture, **Ettercap** has been invoked in its graphical environment. This tool, widely used in network audits, allows the quick visualization of active connections and the unified interception of traffic on the LAN.

Command used:



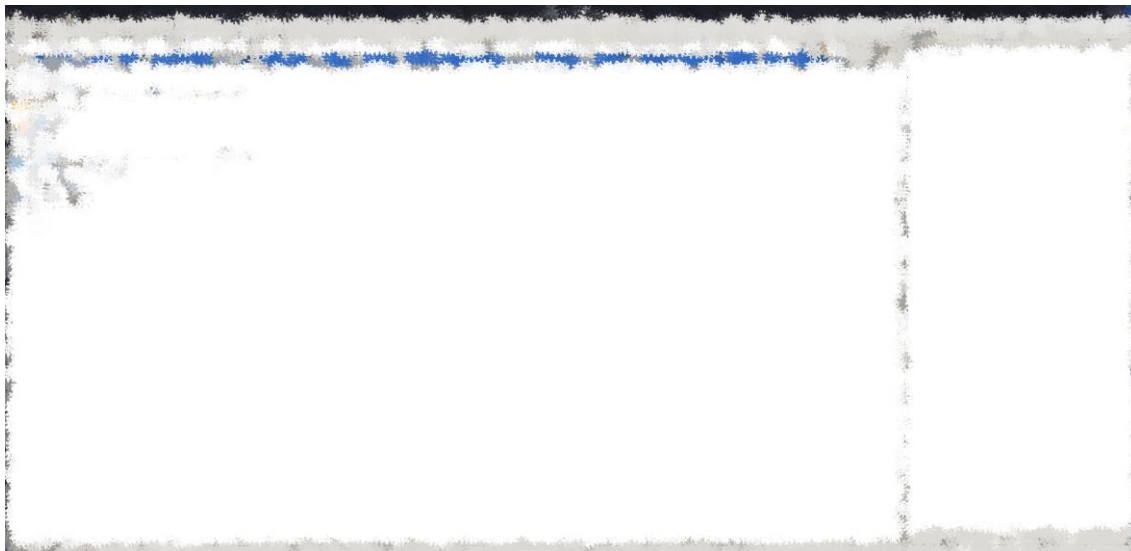
4. Analysis of Captured Information

The effectiveness of *sniffing* is revealed in the dissection phase. The objective was to extract information in clean text, reconstruct sessions and identify protocols.

4.1. Artifact Mining with NetworkMiner

The .pca file generated by tcpdump was imported into **NetworkMiner**, a Network Forensic Analysis (NFAT) tool aimed at automatic data extraction.

Command used:



Analysis of Artifact Extraction with NetworkMiner:

Importing the capture file (lab5_sniffing.pca) into NetworkMiner demonstrated the enormous effectiveness of network forensics. Without generating any type of active traffic that could alert detection systems (IDS), the tool was able to dissect the packets and reconstruct the environment, identifying a total of 23 distinct *hosts* communicating during the capture window.

From the direct observation of the results, three crucial information vectors stand out:

1. **Passive OS Fingerprinting:** NetworkMiner analyzed the packet headers and automatically identified the host operating system [IP Redacted] as Windows, without the need for intrusive scans like those of Nmap.
2. **Navigation Mapping (External Hosts):** The tool isolated communications from the local network to external servers, exposing the contacted domains. Traffic directed to Microsoft infrastructures is visible, namely outlook.office.com (via IPv6) and l.clarity.ms (web telemetry), which allows an auditor to profile the use of the network (e.g., the user opened the email client during the capture).
3. **Metadata Extraction:** The top tabs confirm that in addition to IP mapping, the tool reconstructed 5 full sessions and extracted 34 DNS requests in clean text, underlining the privacy risk inherent in unencrypted traffic on a local network.

5. Conclusions and Mitigation Measures

The execution of Practice Sheet 5 demonstrated, in a controlled environment, the exposure inherent to communications in a local network. From the analysis of the information captured, the following conclusions and defensive perspectives can be drawn:

- **Effectiveness of Passive Capture:** The combination of tcpdump (for stable interception) and NetworkMiner (for artifact processing) has proven to be an extremely capable forensic methodology. It is possible to reconstruct the behavior of a user on the network without generating any noise or alarm (IDS/IPS).
- **Metadata Visibility:** Even in protected communications (such as HTTPS sessions), attributes such as DNS resolutions and certificates exchanged in *TLS handshakes* provide valuable data to an auditor/attacker about the browsing profile and applications in use.
- **Mitigation and Defense:** The primary defense against *sniffing* techniques in local networks is effective network segmentation and rigorous implementation of *end-to-end* encryption. At the corporate infrastructure level, switches must apply *Port Security* rules and ARP Spoofing mitigation (such as *Dynamic ARP Inspection*) to prevent an attacker from being able to expand their visibility beyond their own segment.

6. Command Log

Preparation and Verification:

- `sudo arp -na`

Traffic Capture:

- `sudo tcpdump -i eth0 -s 0 -U -w /tmp/lab5_sniffing.pca -C 500000`
- `ls -lha /tmp/lab5_sniffing.pca`
- `sudo ettercap -G`

Forensic Analysis:

- `sudo mono /opt/NetworkMiner_3-0/NetworkMiner.exe`